What is kube-apiserver?

Professional Answer

The API Server is the central management component that exposes Kubernetes APIs, validates requests, enforces authentication and authorization, and acts as the gateway to the cluster.

Unprofessional Answer

API server talks to Kubernetes.

Why is API Server called the “front door” of Kubernetes?

Professional Answer

All internal and external communication in Kubernetes goes through the API Server, making it the single entry point for cluster interactions.

Unprofessional Answer

Because everything goes through it.

What is kube-apiserver and what does it do?

kube-apiserver is the central management component of Kubernetes that exposes the REST APIs.
All external and internal communication in the cluster goes through the API server.
It handles authentication, authorization, admission control, and request validation.
The API server stores and retrieves cluster state from etcd.
It is stateless and can be horizontally scaled for high availability.

Kubernetes Internals – Component-by-Component

1. kube-apiserver (The Brain Gateway)

What it is internally

- A **stateless REST API service**
- Written in **Go**
- Fronted by HTTPS
- Can be horizontally scaled

Internal Responsibilities

1. **Authentication**
    - Validates *who* is making the request
    - Supports:
        - Certificates

        - Tokens
        - OIDC
        - Service Accounts
2. **Authorization**
    - Decides *what* the user can do
    - Uses:
        - RBAC
        - ABAC
        - Webhooks
3. **Admission Controllers**
    - Intercepts requests **after auth**
    - Can:
        - Validate objects
        - Mutate objects
    - Examples:
        - LimitRanger
        - ResourceQuota
        - PodSecurity
        - MutatingWebhook

4. **API Validation**
    - Validates schema
    - Ensures object correctness
5. **Persistence**
    - Writes state to **etcd**
    - Reads state from etcd

What is kube-apiserver?

The kube-apiserver is the **central management component** of Kubernetes.
It exposes the **Kubernetes API** over REST/HTTPS.
All cluster operations go through it.
It validates, authenticates, authorizes, and persists data.
No other component talks directly to etcd.

Why kube-apiserver is critical

- Single entry point to the cluster
- Controls state transitions
- Enforces security policies
- Mediates between users and control plane
- Failure = control plane outage

If kube-apiserver is down, Kubernetes is down.

Who talks to kube-apiserver?

Component	Interaction
kubectl	REST calls
Scheduler	Watches Pods
Controller Manager	Watches desired vs actual
kubelet	Pod status updates
etcd	Read/write state

Rule: Nothing bypasses the API Server.